

DMBI 2025 May PYQ Answers

Q1. [5 Marks each]

- (a) What is the difference between classification and prediction? Give examples of each.
- (b) What is a Decision Support System (DSS)?
- (c) How can you improve the efficiency of the Apriori algorithm?
- (d) What is K-Means clustering?

Q2. [10 Marks each]

(a) Data Warehousing & OLAP:

- Suppose a data warehouse consists of three dimensions: **time, doctor, and patient**, and two measures: **count and charge**.
- Draw a **star schema diagram** for this data warehouse.
- Starting with the base cuboid [day, doctor, patient], list the specific OLAP operations required to list the total fee collected by each doctor in 2004.

(b) Data Normalization:

- Given a dataset: `[10, 20, 30, 40, 50]`.
- Apply **Min-Max Normalization** (Range [0, 1]) and provide the normalized value for each element.
- Apply **z-score normalization** and provide the normalized value for the input `30`.

Q3. [10 Marks each]

- (a) Explain the following operations with examples: Roll-up, Drill-down, Slice, Dice, and Pivot.
- (b) What is attribute selection in decision trees? Explain **Information Gain** and **Gini Index** with their respective formulas.

Q4. [10 Marks each]

(a) What are ensemble methods, and why are they useful in classification tasks?

(b) What is the BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies) algorithm?

Q5. [10 Marks each]

(a) Apply the Apriori algorithm to the following transactions with **Minimum Support = 60%** and **Confidence = 75%**. Find all frequent itemsets and association rules.

Transaction ID	Items Bought
T1	Milk, Bread
T2	Milk, Diaper
T3	Milk, Bread, Diaper
T4	Bread, Diaper
T5	Milk, Bread

(b) Describe any two outlier detection methods.

Q6. [5 Marks each]

(a) Given scores of 10 students: [50, 55, 58, 60, 65, 70, 72, 75, 80]. Create a **histogram with 3 bins**; show bin ranges and frequency counts in graphical form.

(b) Given the confusion matrix below, calculate **Accuracy** and **Precision**.

	Predicted Yes	Predicted No
Actual Yes	50	10
Actual No	5	35

(c) What are the key differences between **Business Intelligence** and traditional reporting systems?

(d) Write a short note on **Simple Linear Regression**.

Q1. [5 Marks each] - Answers

a. What is the difference between classification and prediction? Give examples of each.

Basis	Classification	Prediction
Definition	Assigns data into predefined classes or categories	Estimates a continuous numeric value
Output Type	Discrete (labels/classes)	Continuous (numeric values)
Goal	Identify class/category of data	Forecast future or unknown values
Techniques Used	Decision Trees, Naive Bayes, KNN, SVM	Regression (Linear, Polynomial), Time Series
Nature of Problem	Categorical problem	Numerical problem
Example	Classifying emails as spam or not spam	Predicting house price based on features
Another Example	Classifying patients as diseased or healthy	Predicting temperature for next day

b. What is a Decision Support System (DSS)?

Definition

A **Decision Support System (DSS)** is a computer-based information system that

helps managers and professionals make better decisions by combining **data, models, and analytical tools**.

Key Points

- **Data Management:** Collects and organizes information from multiple sources.
- **Model Management:** Provides analytical models (e.g., forecasting, optimization).
- **User Interface:** Allows easy interaction for decision-makers.
- **Decision Support:** Assists in solving semi-structured or unstructured problems.
- **Flexibility:** Can adapt to different decision-making scenarios.
- **Integration:** Combines databases, analytical tools, and reporting systems.

Example

- A retail company uses DSS to analyze **sales trends and customer preferences**. Based on this, managers decide which products to stock more during festive seasons.

c. How can you improve the efficiency of the Apriori algorithm?

Definition

The **Apriori algorithm** is used for mining frequent item sets and generating association rules. Its efficiency can be improved by reducing the number of candidate item sets and database scans.

Key Points

- **Transaction Reduction:** Remove transactions that do not contain frequent item sets to minimize scanning.
- **Partitioning:** Divide the database into smaller partitions and mine frequent item sets locally before combining results.
- **Sampling:** Use a random sample of the database to approximate frequent item sets, reducing computation.

- **Dynamic Itemset Counting:** Add new candidate item sets dynamically during database scans instead of generating all at once.
- **Efficient Data Structures:** Use trie or hash trees to store and count item sets more efficiently.

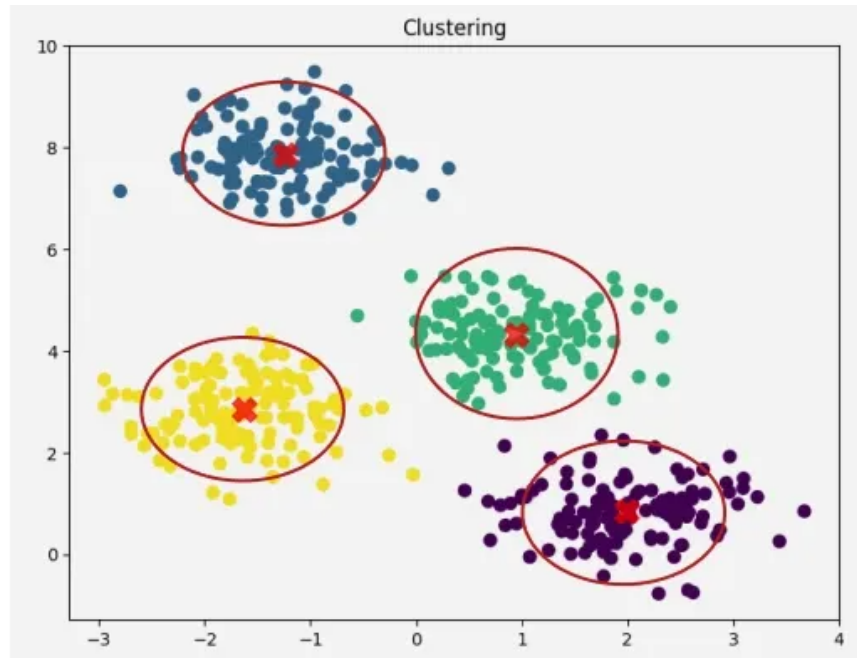
d. What is K-Means clustering?

Definition

K-Means is a **partitioning clustering algorithm** that divides a dataset into k clusters, where each data point belongs to the cluster with the nearest mean (centroid).

Key Points

- **Unsupervised Learning:** Used for grouping unlabeled data.
- **Centroid-Based:** Each cluster is represented by its mean (centroid).
- **Steps:**
 1. Choose number of clusters k .
 2. Initialize k centroids randomly.
 3. Assign each data point to the nearest centroid.
 4. Update centroids as the mean of assigned points.
 5. Repeat until centroids stabilize.
- **Distance Measure:** Usually Euclidean distance is used.
- **Applications:** Market segmentation, image compression, document clustering.

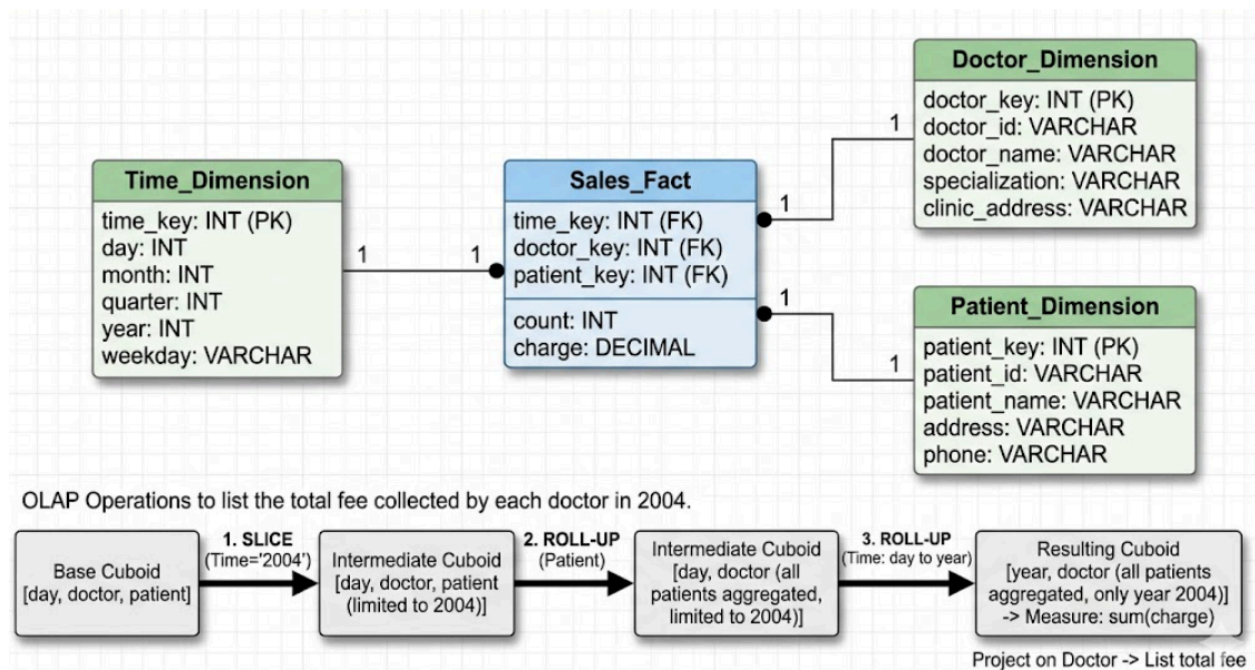


Example

- Suppose we want to group customers into 3 clusters based on **income and spending score**. K-Means will form clusters like:
 - Cluster 1: Low income, low spending
 - Cluster 2: High income, high spending
 - Cluster 3: Moderate income, moderate spending

Q2. [10 Marks each] - Answers

a. Data Warehousing & OLAP: i) Suppose a data warehouse consists of three dimensions: time, doctor, and patient, and two measures: count and charge. ii) Draw a star schema diagram for this data warehouse. iii) Starting with the base cuboid [day, doctor, patient], list the specific OLAP operations required to list the total fee collected by each doctor in 2004.



OLAP Operations

Starting with the base cuboid [day, doctor, patient], the objective is to find the **total fee collected by each doctor in 2004**. This requires the following specific OLAP operations:

1. **Slice** on the **Time** dimension for the value "2004".
2. **Roll-up** on the **Patient** dimension from the **patient** level to the **all** level. (This aggregates, or sums up, the **charge** measure for all patients.)
3. **Roll-up** on the **Time** dimension from the **day** level to the **year** level. (This is necessary because the base cuboid is at the **day** level, and we need data for the full year of 2004.)

The resulting cuboid is [year, doctor], restricted to the year 2004. A simple projection of this cuboid on the **doctor** dimension, showing the **charge** measure, provides the requested list.

- b. Data Normalization:**
- i) Given a dataset: [10, 20, 30, 40, 50].
 - ii) Apply Min-Max Normalization (Range [0, 1]) and provide the

normalized value for each element. iii) Apply z-score normalization and provide the normalized value for the input **30** .

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Q2→b] Given dataset is 10, 20, 30, 40, 50

⇒ Min-Max Normalization = $\frac{x - \min}{\max - \min}$
Formula

Applying the min-max normalization on the above dataset

$$v = \frac{10 - 10}{50 - 10} = 0$$

$$v = \frac{40 - 10}{50 - 10} = \frac{30}{40} = 0.75$$

$$v = \frac{20 - 10}{50 - 10} = \frac{10}{40} = 0.25$$

$$v = \frac{50 - 10}{50 - 10} = \frac{40}{40} = 1$$

$$v = \frac{30 - 10}{50 - 10} = \frac{20}{40} = 0.5$$

Original data (v) = [10, 20, 30, 40, 50]

Ans Normalized data = [0, 0.25, 0.5, 0.75, 1]

⇒ Applying Z-Score Normalization on input 30.

Z-Score Normalization Formula $Z = \frac{x - \mu}{\sigma}$

$$\text{Std. Formula} = \sqrt{\frac{\sum (x_i - \mu)^2}{n - 1}}$$

$$\text{Mean } (\mu) = \frac{10 + 20 + 30 + 40 + 50}{5} = 30$$

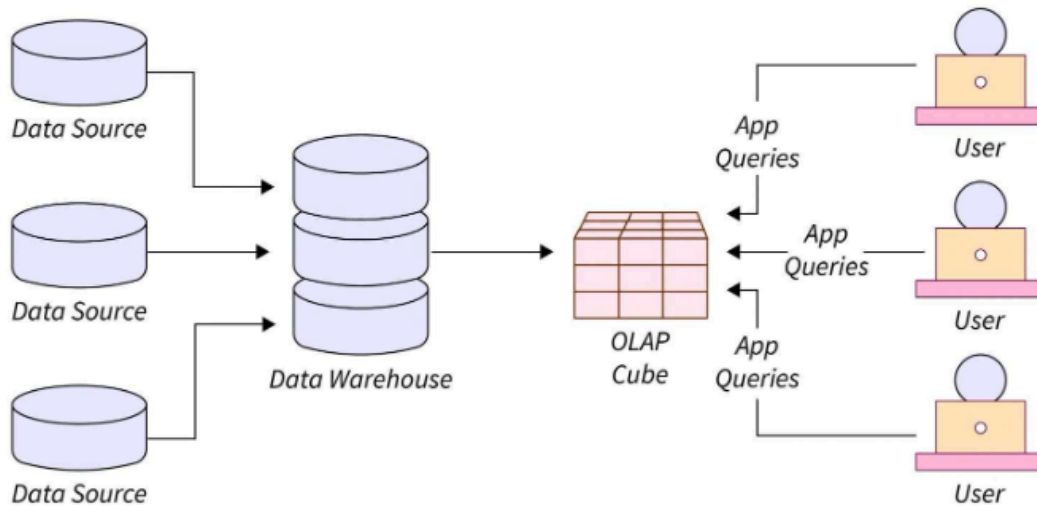
$$\begin{aligned}
 \text{Std} &= \sqrt{\frac{(10-30)^2 + (20-30)^2 + (30-30)^2 + (40-30)^2 + (50-30)^2}{5-1}} \\
 &= \sqrt{\frac{(-20)^2 + (-10)^2 + (10)^2 + (20)^2}{4}} \\
 &= \sqrt{\frac{400 + 100 + 100 + 400}{4}} = \sqrt{\frac{1000}{4}} \\
 &= \sqrt{250} = \boxed{15.81} \\
 \text{Ans } z &= \frac{x - \mu}{\sigma} = \frac{30 - 30}{15.81} = \boxed{0}
 \end{aligned}$$

Q3. [10 Marks each] - Answers

a. Explain the following operations with examples: Roll-up, Drill-down, Slice, Dice, and Pivot.

Definition

OLAP (Online Analytical Processing) operations allow users to analyze multidimensional data from different perspectives. The main operations are **Roll-up**, **Drill-down**, **Slice**, **Dice**, and **Pivot**, which help in summarizing, detailing, filtering, and reorienting data for decision-making.



1. Roll-Up (Aggregation)

- Summarizes data by climbing up a hierarchy or reducing dimensions.
- Example: Viewing *sales data by year* instead of *month*.

2. Drill-Down

- Opposite of roll-up; provides more detailed data by moving down the hierarchy.
- Example: From *year* → *quarter* → *month* → *day*.

3. Slice

- Selects a single dimension of the cube to view a subset of data.
- Example: Sales data for *2025 only*.

4. Dice

- Selects multiple dimensions to form a sub-cube.
- Example: Sales data for *2025* and *Product A* in *Region X*.

5. Pivot (Rotation)

- Reorients the cube to provide a different perspective of data.
- Example: Switching view from *sales by region* to *sales by product*.

b. What is attribute selection in decision trees? Explain Information Gain and Gini Index with their respective formulas.

1. Definition

Attribute selection in decision trees is the process of choosing the best attribute (feature) to split the data at each node.

The goal is to select the attribute that best separates the data into distinct classes.

Attribute Selection in Decision Trees

- At each node, multiple attributes are available for splitting.
- A **splitting criterion** is used to decide the best attribute.
- The best attribute is the one that creates **pure or homogeneous subsets**.
- Common measures:
 - Information Gain (used in ID3)
 - Gini Index (used in CART)

1. Information Gain

- **Concept:** Measures the reduction in uncertainty (entropy) after splitting on an attribute.
- **Formula:**

$$IG(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} \cdot Entropy(S_v)$$

where:

- S = dataset
- A = attribute
- S_v = subset after splitting on value v
- **Entropy Formula:**

$$Entropy(S) = - \sum p_i \log_2(p_i)$$

where p_i = probability of class i .

- **Example:**

- Dataset: "Play Tennis?" with attributes like *Outlook*, *Temperature*, *Humidity*.
- If splitting on **Outlook** reduces entropy the most, it is chosen as the root node.

2. Gini Index

- **Concept:** Measures impurity based on probability of incorrect classification.
- **Formula:**

$$Gini(S) = 1 - \sum p_i^2$$

where p_i = probability of class i .

- **Interpretation:**

- Gini = 0 → Pure node (all samples belong to one class).
- Higher Gini → More impurity.

- **Example:**

- Node with 50% Yes and 50% No → Gini = 0.5 (maximum impurity for binary).
- Node with 100% Yes → Gini = 0 (pure).

Q4. [10 Marks each] - Answers

a. What are ensemble methods, and why are they useful in classification tasks?

1. Definition

Ensemble methods are techniques in machine learning where multiple models (usually classifiers) are combined to improve overall prediction accuracy and performance.

What are Ensemble Methods?

- Instead of using a single model, multiple models are trained.
- Their predictions are combined to produce a final result.
- Idea: **"Many weak learners → one strong learner."**

Types of Ensemble Methods

1. Bagging (Bootstrap Aggregating)

- Multiple models trained on different random samples of data.
- Predictions are combined using voting.
- Reduces **variance**.

Example: Random Forest

2. Boosting

- Models are trained sequentially.
- Each new model focuses on errors of the previous one.
- Improves **accuracy and reduces bias**.

Examples: AdaBoost, Gradient Boosting

3. Voting

- Multiple different models are used.
- Final prediction is based on:
 - Majority voting (classification)

- Averaging (regression)

Why Ensemble Methods are Useful in Classification

- **Improved Accuracy:** Combines strengths of multiple models.
- **Reduced Overfitting:** Especially in bagging methods.
- **Better Generalization:** Performs well on unseen data.
- **Handles Complex Data:** Captures different patterns.
- **Robustness:** Less sensitive to noise and data variations.
- **Error Reduction:** Individual model errors cancel out.

Example

- Instead of one decision tree, a **Random Forest** uses many trees.
- Each tree predicts a class, and the final output is based on majority voting.
- This gives more accurate results than a single tree.

b. What is the BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies) algorithm?

Definition / Introduction

BIRCH (**B**alanced **I**terative **R**educing and **C**lustering using **H**ierarchies) is a clustering algorithm designed for **large datasets**. It incrementally builds a compact summary of the data using a **CF (Clustering Feature) tree**, then applies clustering on this summary. It is efficient, memory-friendly, and handles noise well.

Key Concepts:

1. CF Tree (Clustering Feature Tree)

- A hierarchical data structure that stores compact summaries of data points.

- Each node contains clustering features (number of points, linear sum, square sum).

2. Phases of BIRCH Algorithm

- **Phase 1:** Build CF tree by inserting data points.
- **Phase 2:** Condense the tree by removing outliers/noise.
- **Phase 3:** Apply global clustering (e.g., k-means) on leaf entries.
- **Phase 4 (optional):** Refine clusters further.

3. Advantages

- Handles very large datasets efficiently.
- Requires only one database scan in most cases.
- Robust against noise.

Example

Suppose we have **10,000 customer transaction records**:

- Instead of clustering all 10,000 directly, BIRCH builds a **CF tree** summarizing them into, say, **100 compact nodes**.
- Then clustering (e.g., k-means) is applied on these 100 nodes instead of the full dataset.
- Result: Faster clustering with similar accuracy, even under memory constraints.

Q5. [10 Marks each] - Answers

a. Apply the Apriori algorithm to the following transactions with Minimum Support = 60% and Confidence = 75%. Find all frequent itemsets and association rules.

Step 1: Transaction Table

Transaction ID	Items Bought
T1	Milk, Bread
T2	Milk, Diaper
T3	Milk, Bread, Diaper
T4	Bread, Diaper
T5	Milk, Bread

Step 2: Find Frequent 1-Itemsets (L1)

Candidate 1-itemsets (C1)

Item	Count	Support
Milk	4	$4/5 = 80\%$
Bread	4	$4/5 = 80\%$
Diaper	3	$3/5 = 60\%$

Since all supports are $\geq 60\%$, all are frequent.

Frequent 1-itemsets (L1)

{Milk}, {Bread}, {Diaper}

Step 3: Find Frequent 2-Itemsets (L2)

Candidate 2-itemsets (C2)

Itemset	Count	Support
{Milk, Bread}	3	60%
{Milk, Diaper}	2	40%
{Bread, Diaper}	2	40%

Only itemsets with support $\geq 60\%$ are selected.

Frequent 2-itemsets (L2)

{Milk, Bread}

Step 4: Find Frequent 3-Itemsets (L3)

Candidate 3-itemset

Itemset	Count	Support
{Milk, Bread, Diaper}	1	20%

Support is less than 60%, so rejected.

Frequent 3-itemsets

No frequent 3-itemsets.

Final Frequent Itemsets

Frequent Itemset	Support
{Milk}	80%
{Bread}	80%
{Diaper}	60%
{Milk, Bread}	60%

Step 5: Generate Association Rules

Association rules are generated from frequent itemsets.

From:

{Milk, Bread}

Possible Rules:

1. Milk $\rightarrow$ Bread
2. Bread $\rightarrow$ Milk

Rule 1: Milk → Bread

Confidence Calculation

$$\text{Confidence}(\text{Milk} \rightarrow \text{Bread}) = \frac{\text{Support}(\text{Milk}, \text{Bread})}{\text{Support}(\text{Milk})} = \frac{3}{4} = 75\%$$

Confidence = 75% ✓

Rule Accepted.

Rule 2: Bread → Milk

Confidence Calculation

$$\text{Confidence}(\text{Bread} \rightarrow \text{Milk}) = \frac{\text{Support}(\text{Milk}, \text{Bread})}{\text{Support}(\text{Bread})} = \frac{3}{4} = 75\%$$

Confidence = 75% ✓

Rule Accepted.

Final Association Rules

Association Rule	Confidence
Milk → Bread	75%
Bread → Milk	75%

b. Describe any two outlier detection methods.

Definition

An **outlier** is a data point that significantly differs from other observations in a dataset. Outliers may indicate **errors, rare events, or unusual patterns**, and

detecting them is crucial for improving model accuracy, reliability, and decision-making.

1. Distance-Based Outlier Detection

Concept

- An object is an outlier if it is far away from most other data points.
- Based on distance measures (e.g., Euclidean distance).

Method

- For each data point, compute distance to its neighbors.
- If fewer than a specified number of points lie within a given distance, it is an outlier.

Example

- In a dataset of exam marks (mostly 40–90), a value like 5 or 100 is far from others → outlier.

2. Density-Based Outlier Detection

Concept

- Outliers are points located in **low-density regions** compared to neighbors.
- Uses idea of local density.

Method

- Compare density of a point with density of its neighbors.
- If a point has significantly lower density → outlier.

Example Algorithm

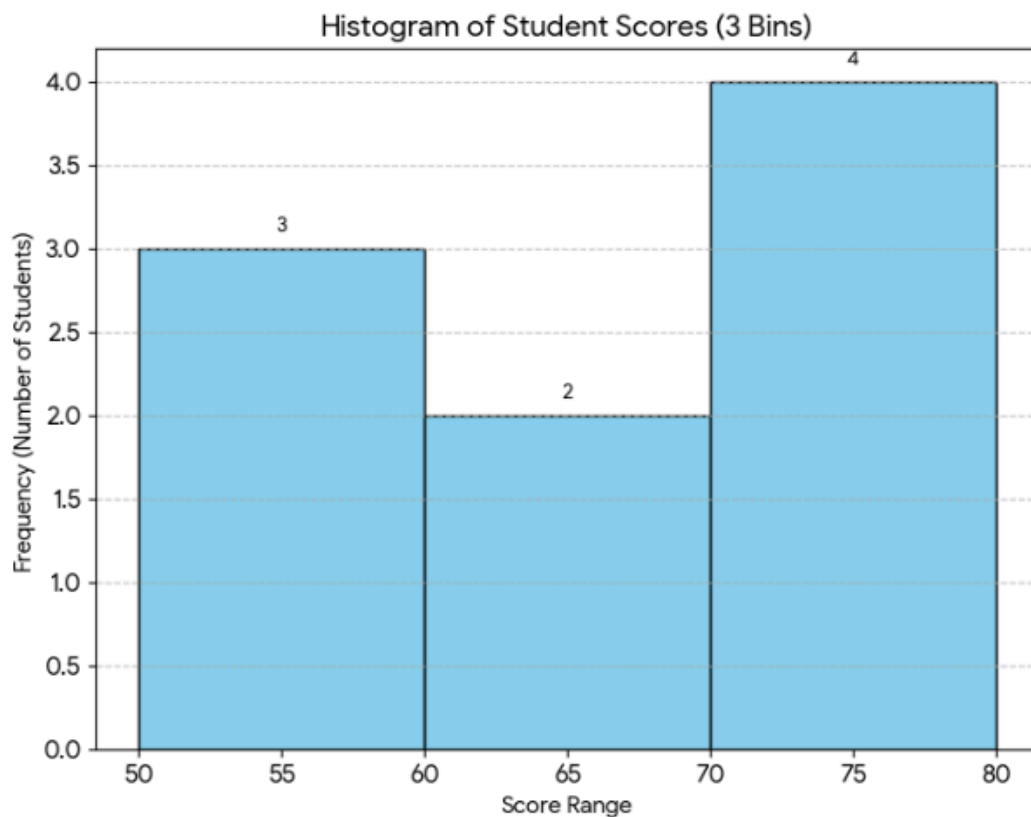
- **LOF (Local Outlier Factor)**

Example

- In a dense cluster of customers, a point isolated from others will have low density → outlier.

Q6. [5 Marks each] - Answers

a. Given scores of 10 students: [50, 55, 58, 60, 65, 70, 72, 75, 80]. Create a histogram with 3 bins; show bin ranges and frequency counts in graphical form.



b. Given the confusion matrix below, calculate Accuracy and Precision.

Confusion Matrix

	Predicted Yes	Predicted No
Actual Yes	50	10
Actual No	5	35

Where:

- TP (True Positive) = 50
- FN (False Negative) = 10
- FP (False Positive) = 5
- TN (True Negative) = 35

1. Accuracy

Accuracy measures how many predictions are correct overall.

Formula:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

Substituting values:

$$Accuracy = \frac{50 + 35}{50 + 35 + 5 + 10}$$

$$Accuracy = \frac{85}{100}$$

$$Accuracy = 0.85 = 85\%$$

$$Accuracy = 85\%$$

2. Precision

Precision measures how many predicted positive cases are actually positive.

Formula:

$$Precision = \frac{TP}{TP+FP}$$

Substituting values:

$$Precision = \frac{50}{50 + 5}$$

$$Precision = \frac{50}{55}$$

$$Precision = 0.909$$

$$Precision \approx 90.9\%$$

$$Precision \approx 90.9\%$$

c. What are the key differences between Business Intelligence and traditional reporting systems?

Basis	Business Intelligence (BI)	Traditional Reporting
Purpose	Supports decision-making with insights	Provides standard reports of data
Data Type	Historical + current, integrated data	Mostly current or transactional data
Analysis	Advanced analysis (trends, patterns, forecasting)	Basic summaries (totals, counts)
Flexibility	Interactive and dynamic (dashboards, drill-down)	Fixed and static reports
Users	Managers, analysts	Operational staff
Tools	OLAP, dashboards, data mining tools	Simple reporting tools

Basis	Business Intelligence (BI)	Traditional Reporting
Decision Support	Strong support for strategic decisions	Limited decision support

Example:

- BI: Dashboard showing sales trends and future predictions
- Traditional Reporting: Monthly sales report showing total sales

DMBI 2025 December PYQ Answers

Q1. [5 Marks each]

- (A) Define different types of attributes with examples.
- (B) Compare OLTP and OLAP operations.
- (C) Define "Business Intelligence" and "Decision Support System" with examples.
- (D) Give the difference between classification and clustering.

Q2. [10 Marks each]

- (A) Explain multidimensional and multilevel association rules with examples.
- (B) Explain the **DBSCAN** algorithm using appropriate diagrams.

Q3. [10 Marks each]

- (A) Explain bagging and boosting in classifiers in brief.
- (B) Use a Hierarchical clustering algorithm to cluster the following 8 points into **3 clusters**: $A1=(2,10)$, $A2=(2,5)$, $A3=(8,4)$, $A4=(5,8)$, $A5=(7,5)$, $A6=(6,4)$, $A7=(1,2)$, $A8=(4,9)$.

Q4. [10 Marks each]

- (A) Explain Market-Basket analysis with an example.
- (B) Describe the BIRCH algorithm in detail.

Q5. [10 Marks each]

- (A) Explain in detail the data mining steps in **Knowledge Discovery in Databases (KDD)** and explain its architecture.
 - (B) Suppose that data for analysis includes the attribute age. The age values for data tuples are (in increasing order):
-

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 .

1. What is the **mean** and **median** of the data?
2. What is the **mode**? Comment on the data's modality.
3. What is the **mid-range**?
4. Give the **five-point summary**.
5. Show the **box plot** of the data.

Q6. [10 Marks each]

(A) Explain the Decision tree classification algorithm.

(B) Explain simple linear Regression with an example.

Q1. [5 Marks each] - Answers

a. Define different types of attributes with examples.

Definition

Attributes are properties or features of data objects. They define the characteristics of the data and determine which statistical or mining techniques can be applied.

Attribute Type	Description	Example
Nominal	Categories without any order	Gender: Male/Female
Ordinal	Ordered categories with ranking	Education Level: High School < Graduate < Postgraduate
Interval	Numeric values with meaningful differences but no true zero	Temperature in °C (20°C – 10°C = 10°C difference)
Ratio	Numeric values with meaningful differences and a true zero	Age, Height, Weight
Binary	Only two possible values	Yes/No, True/False

Example

- **Nominal:** Classifying fruits as *Apple, Mango, Banana*.
- **Ordinal:** Customer satisfaction levels: *Poor, Average, Good, Excellent*.
- **Interval:** Calendar years (2000, 2005, 2010).
- **Ratio:** Salary of employees in dollars.
- **Binary:** Whether a transaction is *Fraudulent (1)* or *Not Fraudulent (0)*.

b. Compare OLTP and OLAP operations. in tabular format

Basis	OLTP	OLAP
Purpose	Manage daily transactions	Analyze data for decision-making
Type of Data	Current, detailed data	Historical, summarized data
Operations	Insert, Update, Delete (CRUD)	Complex queries, analysis
Users	Clerks, operators	Managers, analysts
Query Complexity	Simple queries	Complex queries
Database Design	Normalized	Denormalized (star/snowflake schema)
Response Time	Very fast (real-time)	Slower (due to heavy queries)
Example	Banking transactions, order processing	Sales analysis, business reports

c. Define "Business Intelligence" and "Decision Support System" with examples.

Definition:

Business Intelligence (BI) and Decision Support System (DSS) are systems that help organizations make better decisions using data.

Business Intelligence (BI):

- Refers to technologies and tools used to collect, analyze, and present business data

- Converts raw data into meaningful insights
- Uses dashboards, reports, and data visualization
- Helps in strategic planning and performance analysis

Example:

A company uses BI tools to analyze sales data and identify top-performing products.

Decision Support System (DSS):

- A computer-based system that supports decision-making using data and models
- Helps in solving semi-structured and unstructured problems
- Provides what-if analysis and forecasting
- Supports managers in making informed decisions

Example:

A DSS helps a manager decide the best pricing strategy by analyzing demand and costs.

d. Give the difference between classification and clustering.

Basis	Classification	Clustering
Definition	Assigns data into predefined classes	Groups data based on similarity
Learning Type	Supervised learning	Unsupervised learning
Data Requirement	Requires labeled data	Does not require labeled data
Output	Known class labels	Groups/clusters formed
Goal	Predict class of new data	Discover hidden patterns
Techniques Used	Decision Trees, Naive Bayes, SVM	K-Means, Hierarchical clustering
Example	Classifying emails as spam or not spam	Grouping customers based on behavior

Q2. [10 Marks each] - Answers

a. Explain multidimensional and multilevel association rules with examples.

Definition

Association rules in data mining describe relationships among items in large datasets. Two advanced types are:

- **Multidimensional Association Rules** → involve multiple attributes (dimensions) beyond just items.
- **Multilevel Association Rules** → involve relationships at different levels of abstraction in a hierarchy.

1. Multidimensional Association Rules

- **Concept:** Rules that span across multiple attributes (e.g., age, location, product).
- **Focus:** Not limited to item-item relationships, but include other dimensions.
- **Example:**
 - *"Customers aged 20–30 in Delhi who buy smartphones also buy earphones."*
 - Dimensions: Age, Location, Product.
- **Application:** Marketing, healthcare, customer behavior analysis.

2. Multilevel Association Rules

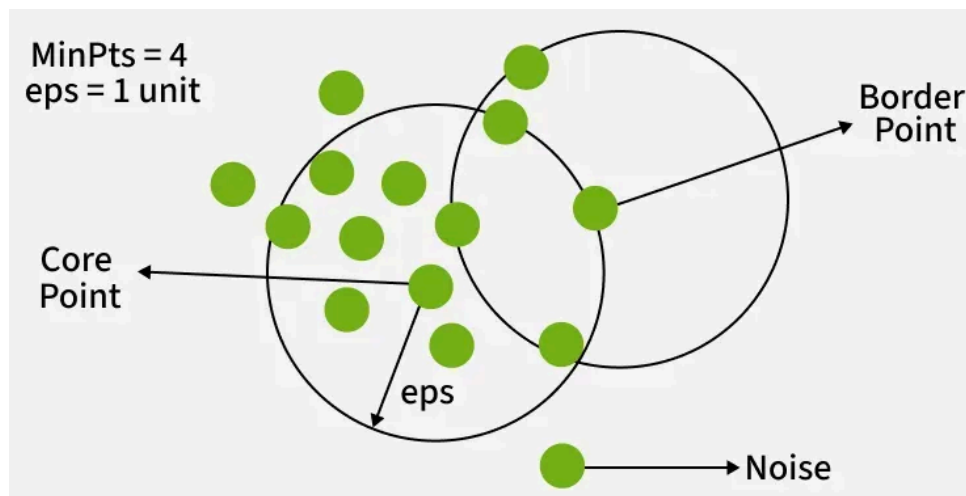
- **Concept:** Rules that explore relationships at different levels of a hierarchy (general → specific).
- **Focus:** Moves from broad categories to detailed subcategories.
- **Example:**

- Higher level: *"If milk is bought, bread is bought."*
- Lower level: *"If Amul milk is bought, Britannia bread is bought."*
- **Application:** Retail product categorization, inventory management.

b. Explain the DBSCAN algorithm using appropriate diagrams.

Definition:

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is an unsupervised clustering algorithm that groups data points based on density and identifies outliers (noise).



Key Points

1. **Core Idea:** Clusters are formed where data points are closely packed together; points in low-density regions are treated as noise.
2. **Parameters:**
 - **ϵ (epsilon)** → Radius of neighborhood around a point.
 - **MinPts** → Minimum number of points required within ϵ to form a dense region.
3. **Types of Points:**
 - **Core Points:** Have at least MinPts neighbors within ϵ .

- **Border Points:** Within ϵ of a core point but not dense enough themselves.
 - **Noise Points:** Neither core nor border, considered outliers.
4. **No Need for Predefined k:** Unlike K-Means, DBSCAN does not require the number of clusters to be specified in advance.
 5. **Advantages:** Handles arbitrary-shaped clusters, detects noise/outliers, and works well with real-world irregular data.
 6. **Limitation:** Sensitive to parameter selection (ϵ and MinPts).

Working Steps of DBSCAN

1. Select an unvisited point.
2. Find all points within ϵ distance (neighborhood).
3. If neighbors $\geq$ MinPts $\rightarrow$ mark as **core point** and form a cluster.
4. Expand cluster by including all density-reachable points.
5. If point is not core $\rightarrow$ mark as **noise** (may later become border).
6. Repeat until all points are processed.

Example

- Suppose we have customer data points plotted by **income vs spending score**.
- DBSCAN with $\epsilon = 3$ and MinPts = 4 may form clusters like:
 - **Cluster 1:** High-income, high-spending customers.
 - **Cluster 2:** Low-income, low-spending customers.
 - **Noise Points:** Irregular customers with unusual spending patterns.

Q3. [10 Marks each] - Answers

a. Explain bagging and boosting in classifiers in brief.

Definition

Bagging and Boosting are **ensemble methods** used in classification to improve accuracy and robustness. Both combine multiple weak learners (often decision trees) but differ in how models are trained and combined.

1. Bagging (Bootstrap Aggregating)

- **Concept:** Builds multiple models independently on different random samples of data.
- **Process:**
 1. Generate bootstrap samples (random subsets with replacement).
 2. Train a classifier (e.g., decision tree) on each sample.
 3. Combine predictions using majority voting (classification).
- **Example: Random Forest** uses bagging with decision trees.
- **Advantage:** Reduces variance and prevents overfitting.

2. Boosting

- **Concept:** Builds models sequentially, each correcting errors of the previous one.
- **Process:**
 1. Train a weak learner on the dataset.
 2. Increase weights of misclassified points.
 3. Train the next learner focusing on difficult cases.
 4. Final prediction is weighted combination of all learners.
- **Example: AdaBoost, Gradient Boosting, XGBoost.**
- **Advantage:** Reduces bias and improves accuracy, especially on complex datasets.

Q4. [10 Marks each] - Answers

a. Explain Market-Basket analysis with an example.

Definition

Market Basket Analysis (MBA) is a **data mining technique** used to discover associations or co-occurrences among items purchased together. It is based on **association rule mining**, which helps businesses understand customer buying behavior.

Key Concepts:

1. **Association Rules:** Express relationships like *"If item A is bought, item B is likely to be bought."*
2. **Support:** Frequency of item sets appearing together.

$$\text{Support}(A \rightarrow B) = \frac{\text{Transactions containing A and B}}{\text{Total transactions}}$$

3. **Confidence:** Probability that item B is bought when item A is bought.

$$\text{Confidence}(A \rightarrow B) = \frac{\text{Transactions containing A and B}}{\text{Transactions containing A}}$$

4. **Lift:** Strength of association compared to random chance.

$$\text{Lift}(A \rightarrow B) = \frac{\text{Confidence}(A \rightarrow B)}{\text{Support}(B)}$$

5. **Applications:** Cross-selling, promotions, product placement, recommendation systems.

Example

- **Retail Store:**
 - Analysis shows customers buying *bread* often also buy *butter*.
 - Rule: *Bread* → *Butter* with high support and confidence.
 - Store places bread and butter close together or offers combo discounts.
- **E-commerce (Amazon/Flipkart):**
 - *"Customers who bought a laptop also bought a laptop bag."*
 - Used for recommendation systems to increase sales.

Q5. [10 Marks each] - Answers

a. Explain in detail the data mining steps in Knowledge Discovery in Databases (KDD) and explain its architecture.

Definition / Introduction

Knowledge Discovery in Databases (**KDD**) is the overall process of extracting meaningful knowledge from large datasets. **Data Mining** is one step within KDD, focusing on applying algorithms to discover patterns. KDD ensures raw data is transformed into valid, useful, and understandable knowledge.

Steps in KDD Process

1. Data Selection

- Choose relevant data from multiple sources (databases, warehouses, files).

2. Data Cleaning & Preprocessing

- Remove noise, handle missing values, correct inconsistencies.

3. Data Transformation / Reduction

- Convert data into suitable formats (normalization, aggregation, dimensionality reduction).

4. Data Mining

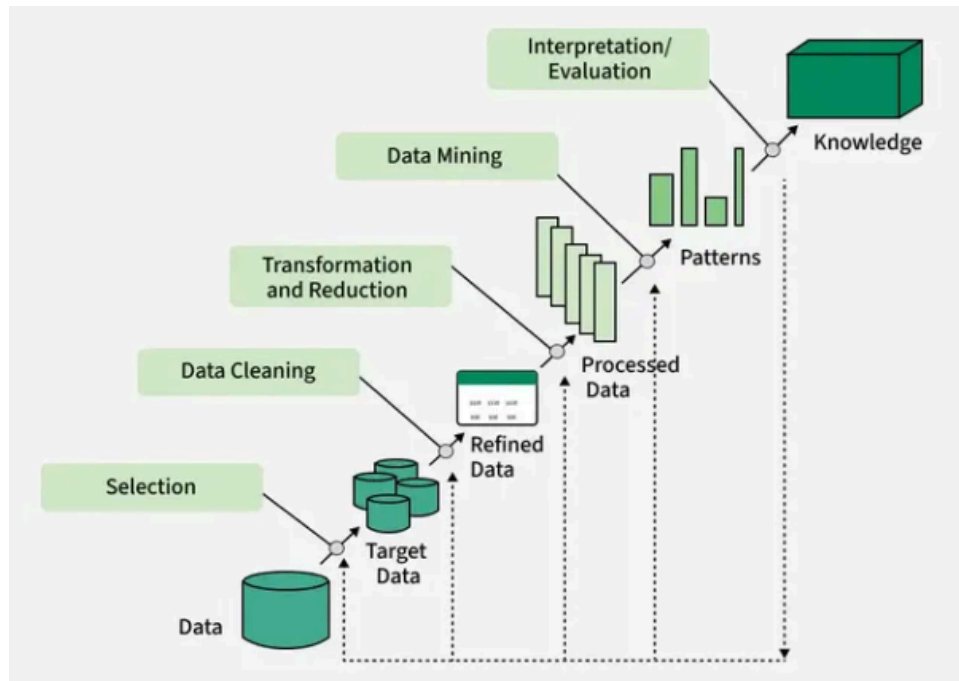
- Apply algorithms (classification, clustering, association rules) to discover patterns.

5. Pattern Evaluation

- Assess discovered patterns for validity, usefulness, and novelty.

6. Knowledge Representation

- Present results using visualization, reports, rules, or charts for decision-making.



KDD Architecture

Layers of KDD Architecture:

1. Data Sources

- Operational databases, flat files, external sources.

2. Data Warehouse

- Central repository integrating data from multiple sources.

3. Preprocessing Layer

- Cleaning, integration, transformation of raw data.

4. Data Mining Engine

- Core algorithms (classification, clustering, regression, association).

5. Pattern Evaluation Module

- Validates interestingness of discovered patterns.

6. Knowledge Presentation Layer

- Visualization tools, reports, dashboards for end-users.

b. Suppose that data for analysis includes the attribute age. The age values for data tuples are (in increasing order): 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 .

Let the given (sorted) data be:

13, 15, 16, 16, 19, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Total values $n=26$

(i) Mean and Median

- **Sum of data = 789**
- **Mean = $789 / 26 = 30.34$ (approx)**
- Since n is even, **Median = average of 13th and 14th values**
 - 13th = 25, 14th = 30
 - **Median = $(25 + 30) / 2 = 27.5$**

(ii) Mode and Modality

- **Most frequent values:** 25 (4 times) and 35 (4 times)
- **Mode = 25 and 35**
- Since there are **two modes**, the data is **bimodal**.

(iii) Mid-range

- **Minimum = 13, Maximum = 70**
- **Mid-range = $(13 + 70) / 2 = 41.5$**

(iv) Five-point Summary

- **Minimum (Min)** = 13
- **Q1 (First Quartile)** = 21
- **Median (Q2)** = 27.5
- **Q3 (Third Quartile)** = 35
- **Maximum (Max)** = 70

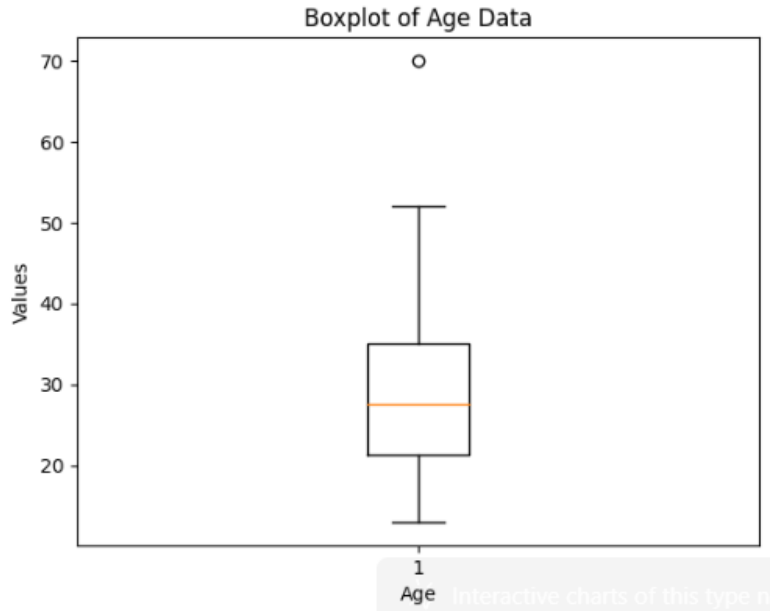
So, **Five-point summary:**

(13, 21, 27.5, 35, 70)

(v) Boxplot (Description)

- Draw a number line from **13 to 70**
- Mark:
 - **Min = 13**
 - **Q1 = 21**
 - **Median = 27.5**
 - **Q3 = 35**
 - **Max = 70**
- Draw a **box** from **21 to 35**, a **line inside** at **27.5**, and **whiskers** from 13 to 70.

Draw the below diagram and not write the above description



Q6. [10 Marks each] - Answers

a. Explain the Decision tree classification algorithm.

Definition

A **Decision Tree** is a supervised classification algorithm that represents decisions in a tree-like structure, where internal nodes denote tests on attributes, branches denote outcomes, and leaf nodes denote class labels.

Structure of Decision Tree

- **Root Node:** Top node representing the best attribute for splitting
- **Internal Nodes:** Decision points based on attributes
- **Branches:** Outcomes of decisions
- **Leaf Nodes:** Final class labels (Yes/No, etc.)

Working of Decision Tree Algorithm

Step 1: Select Best Attribute

- Choose attribute using measures like:

- Information Gain
- Gini Index

Step 2: Split Dataset

- Divide data based on selected attribute values

Step 3: Create Nodes

- Form branches and child nodes

Step 4: Repeat Recursively

- Apply same process to each subset

Step 5: Stop Condition

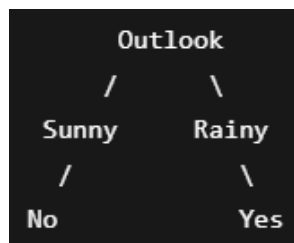
- All data belongs to same class OR
- No attributes left

Example

Dataset: Play Tennis

Outlook	Humidity	Play
Sunny	High	No
Rainy	Normal	Yes

Decision Tree:

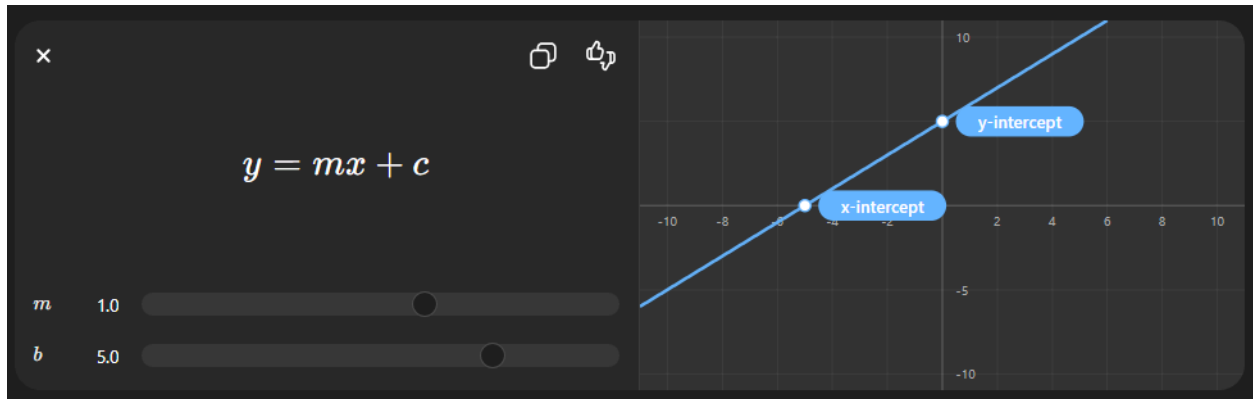


b. Explain simple linear Regression with an example.

Definition:

Simple Linear Regression is a statistical method used to model the relationship between one independent variable (X) and one dependent variable (Y) using a straight line.

Regression Equation



Where:

- y = dependent variable
- x = independent variable
- m = slope (rate of change)
- c = intercept (value of y when $x=0$)

Working of Simple Linear Regression

- Finds the **best-fit line** that minimizes error between actual and predicted values.
- Uses methods like **least squares** to reduce error.
- Predicts output y based on input x .

Steps

1. Collect data (X and Y values)
2. Calculate slope (m) and intercept (c)

3. Form regression equation
4. Use equation to predict values

Example

Suppose:

- X = Hours studied
- Y = Marks

Hours (X)	Marks (Y)
1	40
2	50
3	60

Regression equation:

$$y = 10x + 30$$

- If student studies 4 hours:

$$y = 10(4) + 30 = 70$$

DMBI 2024 May PYQ Answers

Q1. [5 Marks each]

- (A) List out the stages in Data Mining with a neat labeled diagram.
- (B) A sales firm reported the following figures for FY 23-24 (March 23 to Feb 24):
2300, 435, 675, 543, 454, 7877, 5434, 345, 2342, 654, 567, 545 . Show how to normalize this data series using **Min-Max scaling**.
- (C) With an example, explain **Star Versus Snowflake schema** in dimensional modeling.
- (D) What is **Market Basket Analysis**? Explain with a real use case.

Q2. [10 Marks each]

- (A) Draw and list the components of a typical Data Warehouse architecture.
- (B) Consider we have age of 35 participants in a survey given to us in sorted order.
5, 10, 13, 13, 15, 16, 16, 20, 20, 20, 21, 22, 22, 22, 25, 25, 25, 25, 30, 30, 33, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 52, 70, 85 .
Draw histograms for this data taking bin sizes as **5 and 8**. Explain the effect of bin size on the resulting histograms.

Q3. [10 Marks each]

- (A) What is OLAP? Explain various OLAP operations with a neat labeled diagram.
- (B) Explain the working of a decision tree-based classifier. With an example, explain the steps for inducing a tree using the **ID3 algorithm**.

Q4. [10 Marks each]

- (A) Use the Apriori algorithm to identify frequent item-sets in the following database:

Tid	a	b	c	d	e	f	g
Items	1,2,4,5,6	2,3,5	1,2, 4,5	1,2,4, 5	1,2,3,4,5, 6	2,3,4	1,2,4,5

Consider **Minimum Support as 75%** and **Confidence at 85%**. Write down all strong association rules.

(B) What is an outlier? Explain various methods for performing outlier analysis.

Q5. [10 Marks each]

(A) Explain the steps in a hierarchical clustering algorithm. Perform hierarchical clustering on the following 10 points in 2D space: $(2,3)$, $(5,4)$, $(9,6)$, $(4,7)$, $(8,1)$, $(7,2)$, $(6,3)$, $(1,9)$, $(3,6)$, $(4,8)$. Consider that you require **3 clusters**.

(B) Explain mining of **Multilevel** association rules and **Multidimensional** association rules.

Q6. [5 Marks each]

Write short notes on **any four** of the following:

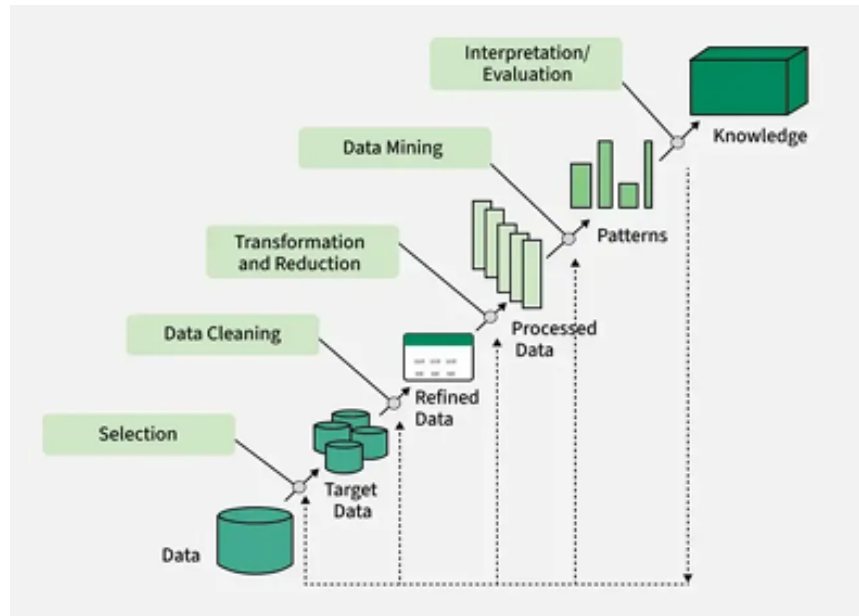
- (A)** Naive Bayes Classifier
- (B)** Bootstrapping
- (C)** BIRCH Algorithm
- (D)** BI Architectures
- (E)** Types of Attributes

Q1. [5 Marks each] - Answers

a. List out the stages in Data Mining with a neat labeled diagram.

Definition

Data Mining is the process of extracting useful patterns and knowledge from large datasets. It involves several systematic stages to ensure meaningful results.



Key Points (Stages)

1. **Data Cleaning** – Remove noise, handle missing values, and correct inconsistencies.
2. **Data Integration** – Combine data from multiple sources into a unified dataset.
3. **Data Selection** – Choose relevant data for the mining task.
4. **Data Transformation** – Convert data into suitable formats (e.g., normalization, aggregation).
5. **Data Mining** – Apply algorithms to discover patterns, associations, or clusters.
6. **Pattern Evaluation** – Assess the discovered patterns for validity and usefulness.
7. **Knowledge Presentation** – Present results using visualization, reports, or rules.

Example:

In a sales system, data is cleaned and combined, analyzed using mining techniques, and results are shown in dashboards for decision-making.

b. A sales firm reported the following figures for FY 23-24 (March 23 to Feb 24): 2300, 435, 675, 543, 454, 7877, 5434, 345, 2342, 654, 567, 545. Show how to normalize this data series using Min-Max scaling.

Min-Max Normalization

Min-Max Normalization is a data transformation technique used to scale values into a fixed range, usually:

[0,1]

Formula for Min-Max Scaling:

$$v' = \frac{v - \min(A)}{\max(A) - \min(A)}$$

Where:

- v = original value
- $\min(A)$ = minimum value in dataset
- $\max(A)$ = maximum value in dataset
- v' = normalized value

Step 1: Given Data

Sales figures:

2300, 435, 675, 543, 454, 7877, 5434, 345, 2342, 654, 567, 545

Step 2: Find Minimum and Maximum Values

- Minimum value:

$\min(A) = 345$

- Maximum value:

$$\max(A) = 7877$$

Step 4: Apply Min-Max Formula

Example Calculations

For 2300

$$\begin{aligned}v' &= \frac{2300 - 345}{7532} \\&= \frac{1955}{7532} \\&\approx 0.260\end{aligned}$$

For 435

$$\begin{aligned}v' &= \frac{435 - 345}{7532} \\&= \frac{90}{7532} \\&\approx 0.012\end{aligned}$$

Step 5: Normalized Data Table

Original Value	Normalized Value
2300	0.260
435	0.012
675	0.044
543	0.026
454	0.014
7877	1.000
5434	0.676

Original Value	Normalized Value
345	0.000
2342	0.265
654	0.041
567	0.029
545	0.027

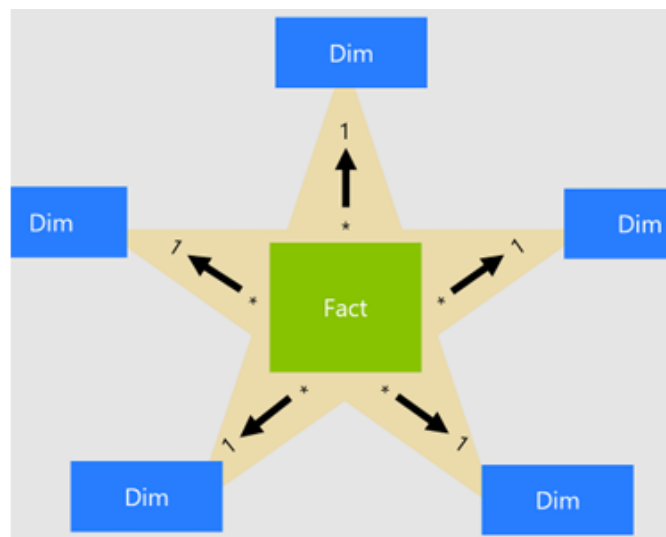
c. With an example, explain Star Versus Snowflake schema in dimensional modeling.

Definition:

Star and Snowflake schemas are data warehouse designs used in dimensional modeling. Both organize data into fact and dimension tables, but differ in structure.

Star Schema:

- Central **fact table** connected directly to multiple **dimension tables**
- Dimension tables are **denormalized** (no further division)
- Simple structure and faster query performance
- Easy to understand and implement



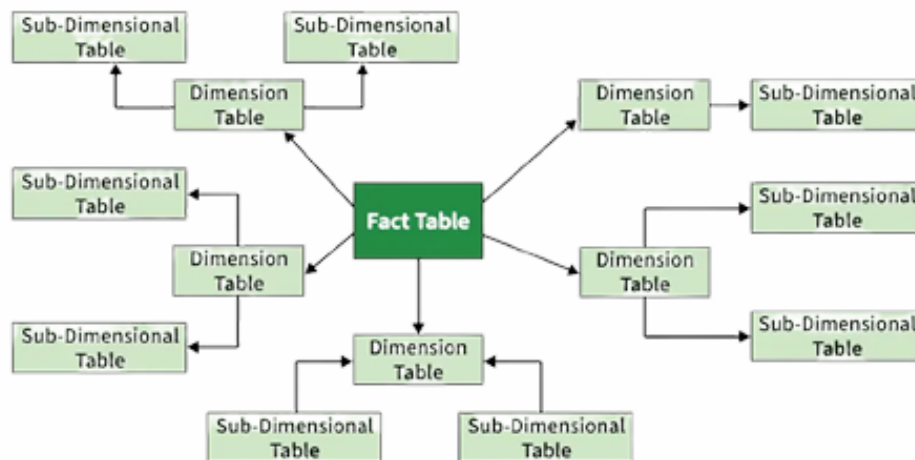
Example:

Fact Table: Sales (Product_ID, Time_ID, Amount)

Dimension Tables: Product, Time, Customer, Location

Snowflake Schema:

- Extension of star schema with **normalized dimension tables**
- Dimension tables are split into multiple related tables
- Reduces data redundancy
- More complex and slightly slower queries due to joins



Example:

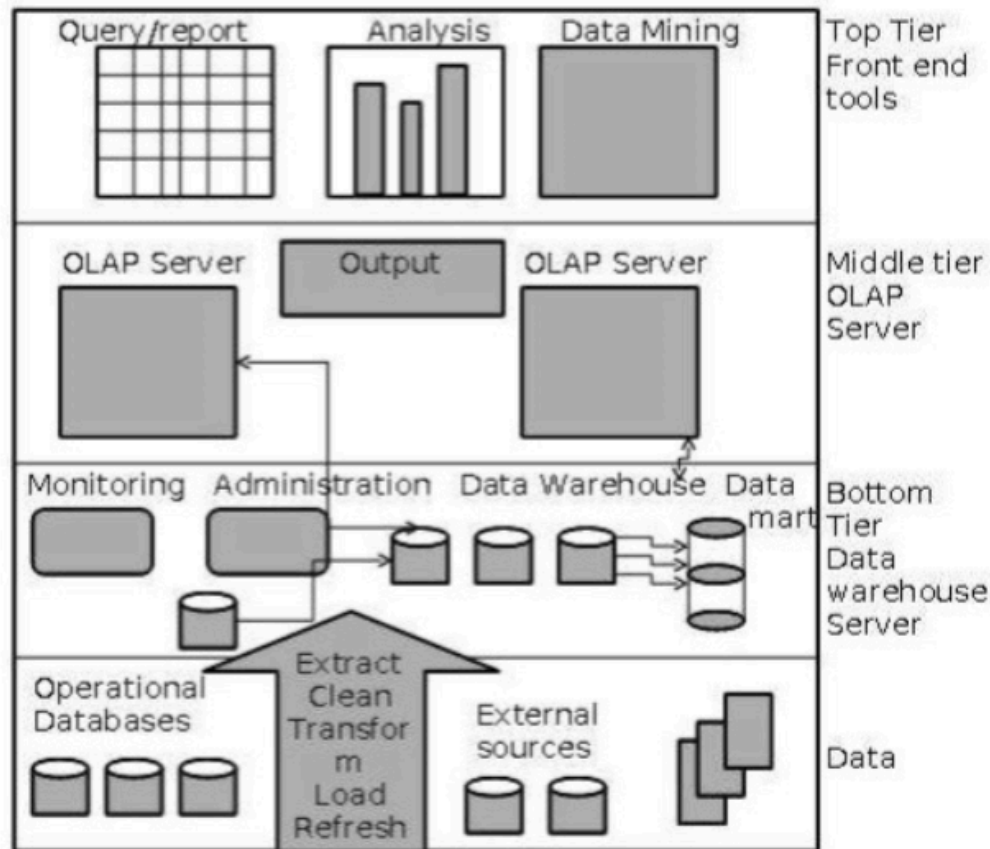
Product dimension split into: Product → Category → Supplier tables

Q2. [10 Marks each] - Answers

a. Draw and list the components of a typical Data Warehouse architecture.

Definition:

Data Warehousing Architecture defines how data is collected, stored, processed, and accessed in a data warehouse for analysis and decision-making.



Main Components / Layers:

- **Data Sources Layer:**
Contains raw data from multiple sources like databases, ERP systems, files, and external sources
- **ETL Layer (Extract, Transform, Load):**
Extracts data from sources, cleans and transforms it, and loads it into the warehouse
- **Data Warehouse Layer:**
Central repository where integrated, historical data is stored
- **Data Mart Layer:**
Subset of data warehouse focused on specific departments (e.g., sales,

finance)

- **OLAP Layer:**

Provides tools for multidimensional analysis (slice, dice, drill-down)

- **Front-End Tools Layer:**

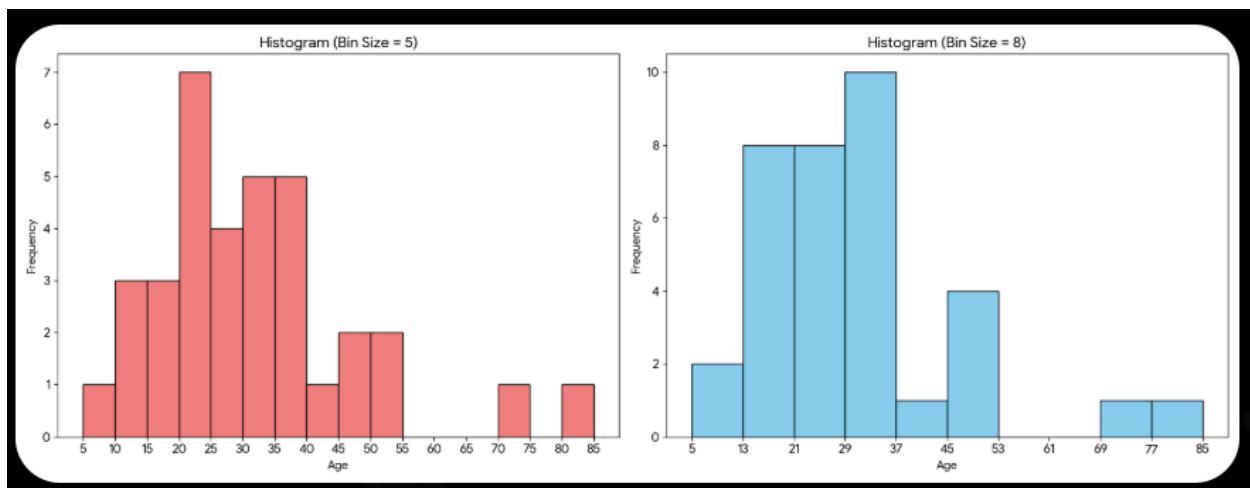
User interfaces like reports, dashboards, and visualization tools for decision-making

Types of Architecture:

- **Single-tier:** Direct access, rarely used
- **Two-tier:** Client directly interacts with warehouse
- **Three-tier:** Most common (Data source → Warehouse → Front-end tools)

b. Consider we have age of 35 participants in a survey given to us in sorted order. 5, 10, 13, 13, 15, 16, 16, 20, 20, 20, 21, 22, 22, 22, 25, 25, 25, 25, 30, 30, 33, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 52, 70, 85 .

Draw histograms for this data taking bin sizes as 5 and 8. Explain the effect of bin size on the resulting histograms.



Effect of Bin Size on Histograms

Choosing the right bin size is a balancing act between **granularity** and **smoothness**. Here is how the different sizes affect the interpretation of the survey data:

Smaller Bin Size (e.g., 5)

- **Granularity:** Provides a more detailed view of the data. You can see specific clusters, such as the peak between ages 20 and 25.
- **Noise:** It is more susceptible to "noise." If the dataset were smaller, a small bin size might make the distribution look jagged or disconnected, even if the underlying trend is smooth.
- **Gaps:** It highlights empty intervals more clearly (e.g., the gap between ages 55 and 70).

Larger Bin Size (e.g., 8)

- **Smoothing:** It "smooths out" the data by grouping more values together. This makes it easier to see the overall shape of the distribution (in this case, it's roughly unimodal and right-skewed).
- **Loss of Detail:** Specific local peaks are hidden. For instance, the distinct concentration of participants in their early 20s vs. early 30s becomes a single large block.
- **Generalization:** Better for identifying the "big picture" or general trends when dealing with very large or highly variable datasets.

Q3. [10 Marks each] - Answers

b. Explain the working of a decision tree-based classifier. With an example, explain the steps for inducing a tree using the ID3 algorithm.

Definition

A **Decision Tree Classifier** is a supervised learning algorithm that classifies data by recursively splitting it based on attribute values. Each internal node represents an attribute test, branches represent outcomes, and leaf nodes represent class

labels. The goal is to maximize classification accuracy by selecting attributes that best separate the data.

Working of Decision Tree Classifier

1. **Start with Dataset** → Contains features and target labels.
2. **Attribute Selection** → Choose the best attribute using measures like **Information Gain** or **Gini Index**.
3. **Split Data** → Partition dataset based on attribute values.
4. **Recursive Partitioning** → Repeat attribute selection and splitting for each subset.
5. **Stopping Criteria** → Stop when:
 - All samples in a node belong to one class, or
 - No attributes remain.
6. **Leaf Nodes** → Represent final class labels (decision outcomes).

ID3 Algorithm (Induction of Decision Tree)

Steps:

1. **Calculate Entropy** of the dataset:

$$Entropy(S) = - \sum p_i \log_2(p_i)$$

where p_i = probability of class i .

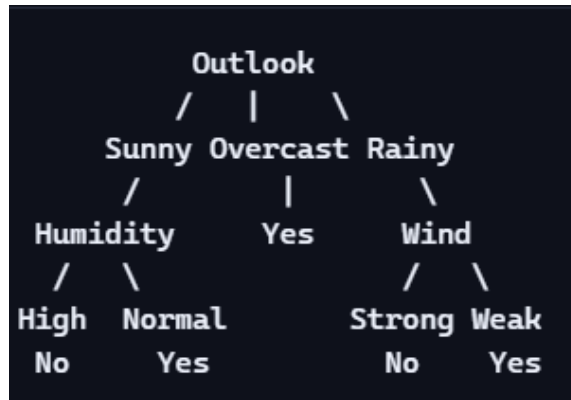
2. **Compute Information Gain** for each attribute:

$$IG(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} \cdot Entropy(S_v)$$

3. **Select Attribute with Highest Information Gain** → becomes root node.
4. **Split Dataset** based on chosen attribute's values.
5. **Repeat Recursively** until all instances are classified or no attributes remain.

Example (Weather Dataset – “Play Tennis?”)

- Attributes: Outlook, Temperature, Humidity, Wind.
- Step 1: Calculate entropy of target (Play Tennis: Yes/No).
- Step 2: Compute information gain for each attribute.
- Step 3: Suppose **Outlook** has highest gain → becomes root.
- Step 4: Split into Sunny, Overcast, Rainy branches.
- Step 5: Continue recursively until leaves classify “Yes” or “No”.



Q4. [10 Marks each] - Answers

a. Use the Apriori algorithm to identify frequent item-sets in the following database. Consider Minimum Support as 75% and Confidence at 85%. Write down all strong association rules.

Step 1: Transaction Database

Tid	Items
a	1, 2, 4, 5, 6
b	2, 3, 5
c	1, 2, 4, 5
d	1, 2, 4, 5
e	1, 2, 3, 4, 5, 6
f	2, 3, 4
g	1, 2, 4, 5

Step 2: Minimum Support Count

$$\begin{aligned}\text{Minimum Support Count} &= 75\% \times 7 \\ &= 0.75 \times 7 = 5.25\end{aligned}$$

Therefore,

Minimum Support Count = 6



Step 3: Find Frequent 1-Itemsets (L1)

Candidate 1-itemsets (C1)

Item	Count	Support
1	5	71.4%
2	7	100%
3	3	42.8%
4	6	85.7%
5	6	85.7%
6	2	28.5%

Only items with count ≥ 6 are frequent.

Frequent 1-itemsets (L1)

{2}, {4}, {5}

Step 4: Find Frequent 2-Itemsets (L2)

Candidate 2-itemsets

Itemset	Count	Support
{2,4}	6	85.7%

Itemset	Count	Support
{2,5}	6	85.7%
{4,5}	5	71.4%

Only itemsets with support $\geq 75\%$ survive.

Frequent 2-itemsets (L2)

{2,4}, {2,5}

Step 5: Find Frequent 3-Itemsets (L3)

Candidate 3-itemset

Itemset	Count	Support
{2,4,5}	5	71.4%

Support $< 75\%$, therefore rejected.

No frequent 3-itemsets.

Final Frequent Itemsets

Frequent Itemset	Support
{2}	100%
{4}	85.7%
{5}	85.7%
{2,4}	85.7%
{2,5}	85.7%

Step 6: Generate Association Rules

Association rules are generated from frequent itemsets.

Rules from {2,4}

Rule 1: $2 \rightarrow 4$

Confidence:

$$Confidence(2 \rightarrow 4) = \frac{Support(2,4)}{Support(2)} = \frac{6}{7} = 85.7\%$$

Confidence = 85.7% ✓

Strong Rule.

Rule 2: $4 \rightarrow 2$

Confidence:

$$Confidence(4 \rightarrow 2) = \frac{Support(2,4)}{Support(4)} = \frac{6}{6} = 100\%$$

Confidence = 100% ✓

Strong Rule.

Rules from {2,5}

Rule 3: $2 \rightarrow 5$

Confidence:

$$\text{Confidence}(2 \rightarrow 5) = \frac{\text{Support}(2,5)}{\text{Support}(2)} = \frac{6}{7} = 85.7\%$$

Confidence = 85.7% ✓

Strong Rule.

Rule 4: $5 \rightarrow 2$

Confidence:

$$\text{Confidence}(5 \rightarrow 2) = \frac{\text{Support}(2,5)}{\text{Support}(5)} = \frac{6}{6} = 100\%$$

Confidence = 100% ✓

Strong Rule.

Final Strong Association Rules

Association Rule	Confidence
$2 \rightarrow 4$	85.7%
$4 \rightarrow 2$	100%
$2 \rightarrow 5$	85.7%
$5 \rightarrow 2$	100%

b. What is an outlier? Explain various methods for performing outlier analysis.

Definition

An **outlier** is a data point that significantly differs from other observations in a dataset. Outliers may indicate **errors**, **rare events**, or **unusual patterns**, and

detecting them is crucial for improving model accuracy, reliability, and decision-making.

Methods for Outlier Analysis

1. Statistical Methods

- Based on distribution properties (mean, variance, z-scores).
- Rule: A point is an outlier if it lies beyond a threshold (e.g., 2 or 3 standard deviations from the mean).
- Example: Exam scores mostly between 40–80, but one student scores 5 → flagged as outlier.

2. Distance-Based Methods

- Outliers are points far away from others in terms of distance (Euclidean, Manhattan).
- Rule: A point is an outlier if fewer than k neighbors lie within radius r .
- Example: Customer spends ₹50,000 while most spend ₹500–₹2000 → flagged as outlier.

3. Density-Based Methods

- Outliers lie in low-density regions compared to dense clusters.
- Algorithm: **DBSCAN** marks points not belonging to any cluster as noise.
- Example: In spatial data, isolated points far from dense groups are outliers.

4. Clustering-Based Methods

- Use clustering algorithms (e.g., k-means).
- Points that do not fit well into any cluster or lie far from cluster centroids are considered outliers.
- Example: In customer segmentation, a buyer with unusual purchasing behavior may not belong to any cluster.

Q6. [10 Marks each] - Answers

a. Write short notes on: Naive Bayes Classifier

Definition:

Naive Bayes Classifier is a probabilistic classification algorithm based on **Bayes' Theorem**, which assumes that all features are independent of each other.

Key Points

- **Bayes' Theorem:**

$$P(C|X) = \frac{P(X|C) \cdot P(C)}{P(X)}$$

- $P(C|X)$: Probability of class C given data X .
- $P(X|C)$: Likelihood of data given class.
- $P(C)$: Prior probability of class.
- $P(X)$: Evidence (normalization factor).

Key Points:

- Uses probability to predict the class of data
- Assumes **feature independence** (naive assumption)
- Works well with large datasets and text data
- Fast and simple to implement
- Performs well even with limited training data
- Commonly used in **spam detection, sentiment analysis**.

Example:

Classifying an email as **spam or not spam** based on words present in the email.

b. Write short notes on: Bootstrapping

Definition:

Bootstrapping is a statistical resampling technique used to estimate the accuracy of a model by repeatedly sampling data **with replacement**.

Key Points:

- Creates multiple samples (bootstrap samples) from the original dataset
 - Each sample is of the same size as the original data
 - Some data points may appear multiple times, others may be left out
 - Used to estimate **model accuracy, variance, and confidence intervals**
 - Helps when dataset is small
 - Commonly used in **ensemble methods** like bagging
 - Improves stability and reliability of models
-

Example:

From a dataset of 100 records, multiple new samples of 100 records are created by random selection with replacement, and a model is trained on each sample to evaluate performance.

DMBI 2024 December PYQ Answers

Q1. [5 Marks each]

(A) Give the difference between **OLAP** and **OLTP**.

(B) Explain Confusion Matrix. Calculate **Accuracy, Precision, and Recall** for the following Cancer Classes Confusion Matrix:

Cancer Classes	Yes	No	Total
Yes	90	210	300
No	140	9560	9700
Total	230	9770	10000

(C) Explain Multilevel and Multidimensional Association rules with suitable example.

(D) Explain the **DBSCAN algorithm** with an example.

Q2. [10 Marks each]

(A) What is **noisy data**? How to handle it?

For the following data $D = \{4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34\}$ and Number of bins = 3

perform the following:

1. Partition into equal frequency bins.
2. Smoothing by bin means.
3. Smoothing by bin boundaries.

(B) Explain the K-means algorithm in detail. Apply K-means algorithm to divide the set of values $\{2, 3, 6, 8, 9, 12, 15, 18, 22\}$ into **3 clusters**.

Q3. [10 Marks each]

(A) **Explain** different types of attributes used in data exploration with examples.

(B) Using the training dataset below, classify the tuple using Naive Bayes

Algorithm: <Homeowner: No, Marital Status: Married, Job experience: 3>

Homeowner	Marital Status	Job experience (in years)	Defaulted
Yes	Single	3	No
No	Married	4	No
No	Single	5	No
Yes	Married	4	No
No	Divorced	2	Yes
No	Married	4	No
Yes	Divorced	2	No
No	Married	3	Yes
No	Married	3	No
Yes	Single	2	Yes

Q4. [10 Marks each]

(A) What do you mean by data mining? Explain the **KDD process** with the help of a suitable diagram.

(B) Apply the Apriori algorithm to the table below (Min. Support 30%, Min. Confidence 70%) to show frequent itemsets and strong association rules.

TID	Items
01	1,3,4,6
02	2,3,5,7
03	1,2,3,5,8
04	2,5,9,10
05	1,4

Q5. [10 Marks each]

(A) **Design** a BI system for **Fraud Detection**. Explain all steps from data collection to decision making.

(B) Compare **Star schema** and **Snowflake schema**.

Q6. [10 Marks each]

(A) What is an **outlier**? List types of outliers and describe methods used for outlier analysis.

(B) Describe different steps involved in **data preprocessing**.

(C) Explain **Bagging** of a classifier.

Q1. [5 Marks each] - Answers

b. Explain Confusion Matrix. Calculate Accuracy, Precision, and Recall for the following Cancer Classes Confusion Matrix:

Definition:

A **Confusion Matrix** is a performance evaluation tool for classification models. It compares the **actual labels** with the **predicted labels** to show how well the model performs.

Structure of Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Key Terms

- **True Positive (TP):** Correctly predicted positive cases.
- **True Negative (TN):** Correctly predicted negative cases.
- **False Positive (FP):** Incorrectly predicted positive (Type I error).
- **False Negative (FN):** Incorrectly predicted negative (Type II error).

Confusion Matrix

	Yes (Predicted)	No (Predicted)	Total
Yes (Actual)	90	210	300

	Yes (Predicted)	No (Predicted)	Total
No (Actual)	140	9560	9700
Total	230	9770	10000

Where:

- **TP (True Positive) = 90**
- **FN (False Negative) = 210**
- **FP (False Positive) = 140**
- **TN (True Negative) = 9560**

1. Accuracy

Accuracy measures the overall correctness of the model.

Formula:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

Substituting values:

$$\begin{aligned}
 Accuracy &= \frac{90 + 9560}{90 + 9560 + 140 + 210} \\
 &= \frac{9650}{10000} \\
 &= 0.965 \\
 &= 96.5\%
 \end{aligned}$$

Accuracy = 96.5%

2. Precision

Precision measures how many predicted positive cases are actually positive.

Formula:

$$Precision = \frac{TP}{TP+FP}$$

Substituting values:

$$\begin{aligned} Precision &= \frac{90}{90 + 140} \\ &= \frac{90}{230} \\ &= 0.3913 \\ &\approx 39.13\% \end{aligned}$$

Precision $\approx$ 39.13%

3. Recall

Recall measures how many actual positive cases are correctly identified.

Formula:

$$Recall = \frac{TP}{TP+FN}$$

Substituting values:

$$\begin{aligned} Recall &= \frac{90}{90 + 210} \\ &= \frac{90}{300} \\ &= 0.30 \\ &= 30\% \end{aligned}$$

Recall = 30%

Conclusion

Measure	Value
Accuracy	96.5%
Precision	39.13%
Recall	30%

Q2. [10 Marks each] - Answers

a. What is noisy data? How to handle it? For the following data D = {4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34} and Number of bins = 3. perform the following:

1. Partition into equal frequency bins.

2. Smoothing by bin means.

3. Smoothing by bin boundaries.

Noisy Data answer is in PYQ of 2023 May (Q5 → b)

Given Data: $D = \{4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34\}$

Number of bins = 3

Total data values = 12

Bin size = $12 / 3 = 4$

So, each bin contains 4 values.

1. Partition into Equal Frequency Bins

Bin 1: $\{4, 8, 9, 15\}$

Bin 2: $\{21, 21, 24, 25\}$

Bin 3: $\{26, 28, 29, 34\}$

2. Smoothing by Bin Means

In this method, every value in a bin is replaced by the mean of that bin.

Bin 1 Mean

Mean = $(4 + 8 + 9 + 15) / 4 = 36 / 4 = 9$

Smoothed Bin: $\{9, 9, 9, 9\}$

Bin 2 Mean

Mean = $(21 + 21 + 24 + 25) / 4 = 91 / 4 = 22.75$

Smoothed Bin: $\{22.75, 22.75, 22.75, 22.75\}$

Bin 3 Mean

Mean = $(26 + 28 + 29 + 34) / 4 = 117 / 4 = 29.25$

Smoothed Bin: $\{29.25, 29.25, 29.25, 29.25\}$

Final Data after Bin Means Smoothing

{9, 9, 9, 9, 22.75, 22.75, 22.75, 22.75, 29.25, 29.25, 29.25, 29.25}

3. Smoothing by Bin Boundaries

In this method, each value is replaced by the nearest boundary value of the bin.

Bin 1

Original Bin: {4, 8, 9, 15}

Boundaries: 4 and 15

Smoothed Values:

4 → 4

8 → 4

9 → 15

15 → 15

Smoothed Bin: {4, 4, 15, 15}

Bin 2

Original Bin: {21, 21, 24, 25}

Boundaries: 21 and 25

Smoothed Values:

21 → 21

21 → 21

24 → 25

25 → 25

Smoothed Bin: {21, 21, 25, 25}

Bin 3

Original Bin: {26, 28, 29, 34}

Boundaries: 26 and 34

Smoothed Values:

26 → 26

28 → 26

29 → 26

34 → 34

Smoothed Bin: {26, 26, 26, 34}

Final Data after Boundary Smoothing

{4, 4, 15, 15, 21, 21, 25, 25, 26, 26, 26, 34}

b. Explain the K-means algorithm in detail. Apply K-means algorithm to divide the set of values {2, 3, 6, 8, 9, 12, 15, 18, 22} into 3 clusters.

Definition

The **K-Means Algorithm** is a **partitioning clustering method** that divides a dataset into k clusters, where each cluster is represented by its centroid (mean). It is an **unsupervised learning technique** widely used in data mining.

Steps in K-Means Algorithm

1. **Choose number of clusters (k).**
2. **Initialize centroids** randomly (or by heuristic).
3. **Assign data points** to the nearest centroid (using Euclidean distance).
4. **Update centroids** by calculating the mean of points in each cluster.
5. **Repeat steps 3–4** until centroids stabilize (no change).

Application to Dataset

Dataset: {2, 3, 6, 8, 9, 12, 15, 18, 22}

Number of clusters (k): 3

Step 1: Initialization

Let's choose initial centroids as: 2, 9, 18

Step 2: Assign points to nearest centroid

- Cluster 1 (centroid = 2): {2, 3, 6}
- Cluster 2 (centroid = 9): {8, 9, 12}
- Cluster 3 (centroid = 18): {15, 18, 22}

Step 3: Update centroids

- Cluster 1 mean = $(2+3+6)/3 = 3.67 \approx 4$
- Cluster 2 mean = $(8+9+12)/3 = 9.67 \approx 10$
- Cluster 3 mean = $(15+18+22)/3 = 18.33 \approx 18$

Step 4: Reassign points (stable clusters)

- Cluster 1: {2, 3, 6} → centroid ≈ 4
- Cluster 2: {8, 9, 12} → centroid ≈ 10
- Cluster 3: {15, 18, 22} → centroid ≈ 18

Final Clusters

- **Cluster 1:** {2, 3, 6}
- **Cluster 2:** {8, 9, 12}
- **Cluster 3:** {15, 18, 22}

Q3. [10 Marks each] - Answers

b. Using the training dataset below, classify the tuple using Naive Bayes Algorithm: **<Homeowner: No, Marital Status: Married, Job experience: 3>**

We classify the tuple: **(Homeowner = No, Marital = Married, JobExp = 3)** using Naïve Bayes.

Step 1: Prior Probabilities

From the table (10 records):

- **P(Default = Yes) = 3 / 10** (records: 5, 8, 10)
- **P(Default = No) = 7 / 10**

Step 2: Conditional Probabilities

2.1 For Default = Yes (3 records)

Records with Default = Yes:

(No, Divorced, 2), (No, Married, 3), (Yes, Single, 3)

- $P(\text{Homeowner} = \text{No} \mid \text{Yes}) = 2 / 3$
- $P(\text{Marital} = \text{Married} \mid \text{Yes}) = 1 / 3$
- $P(\text{JobExp} = 3 \mid \text{Yes}) = 2 / 3$

So,

$$P(X|Yes) = \frac{2}{3} \times \frac{1}{3} \times \frac{2}{3}$$

2.2 For Default = No (7 records)

Records with Default = No = 7

- $P(\text{Homeowner} = \text{No} \mid \text{No}) = 4 / 7$
- $P(\text{Marital} = \text{Married} \mid \text{No}) = 3 / 7$
- $P(\text{JobExp} = 3 \mid \text{No}) = 2 / 7$

So,

$$P(X|No) = \frac{4}{7} \times \frac{3}{7} \times \frac{2}{7}$$

Step 3: Compute Posterior Probabilities (Proportional)

$$P(Yes|X) \propto P(Yes) \times P(X|Yes) = \frac{3}{10} \times \frac{2}{3} \times \frac{1}{3} \times \frac{2}{3}$$

$$P(No|X) \propto P(No) \times P(X|No) = \frac{7}{10} \times \frac{4}{7} \times \frac{3}{7} \times \frac{2}{7}$$

Now compare:

- $P(Yes|X) \propto \frac{3}{10} \times \frac{4}{27} = \frac{12}{270} = \frac{2}{45} \approx 0.044$
- $P(No|X) \propto \frac{7}{10} \times \frac{24}{343} = \frac{168}{3430} \approx 0.049$

Step 4: Decision

Since

$$P(No|X) > P(Yes|X)$$

Final Classification:

The given tuple is classified as: **Default = No**

(Homeowner = No, Marital = Married, JobExp = 3) → Default = No

Q5. [10 Marks each] - Answers

a. Design a BI system for Fraud Detection. Explain all steps from data collection to decision making.

Definition

A **Business Intelligence (BI) system for Fraud Detection** integrates data mining, analytics, and visualization to identify suspicious activities. It transforms raw transactional data into actionable insights, helping organizations detect and prevent fraud effectively.

Steps in BI System for Fraud Detection

1. Data Collection

- Gather data from multiple sources:
 - Transaction logs (banking, e-commerce, insurance).
 - Customer profiles (age, location, history).
 - External data (credit scores, blacklists).

2. Data Cleaning & Integration

- Remove duplicates, handle missing values, and standardize formats.
- Integrate data into a **central data warehouse** for consistency.

3. Data Transformation

- Normalize values (e.g., transaction amounts).
- Create derived attributes (e.g., average transaction per day).
- Reduce dimensionality for efficiency.

4. Data Mining Techniques

- **Classification:** Train models to classify transactions as "fraud" or "legitimate."
- **Clustering:** Group similar transactions; outliers may indicate fraud.
- **Association Rules:** Detect unusual item combinations (e.g., sudden high-value purchases).
- **Anomaly Detection:** Identify deviations from normal customer behavior.

5. Pattern Evaluation

- Validate discovered fraud patterns using metrics like precision, recall, and lift.
- Filter out false positives to avoid unnecessary alerts.

6. Knowledge Representation

- Dashboards showing fraud trends, suspicious accounts, and high-risk regions.
- Alerts and reports for fraud analysts.

7. Decision Making

- Managers and fraud analysts use BI insights to:
 - Block suspicious transactions.
 - Investigate flagged accounts.
 - Strengthen fraud prevention policies.

Example

- **Banking Scenario:**
 - A BI system detects that a customer usually spends ₹5,000 monthly but suddenly makes a ₹2,00,000 international transfer.
 - The anomaly is flagged, analysts review the case, and the transaction is temporarily blocked pending verification.

Q6. [10 Marks each] - Answers

b. Describe different steps involved in data preprocessing.

Data preprocessing is the essential step in data mining and machine learning where raw data is prepared for analysis. It ensures that the dataset is accurate, consistent, and suitable for building models. The different steps involved are:



1. Data Cleaning

- Handle missing values, remove duplicates, and correct inconsistencies.
- Example: Filling missing age values with the average age.

2. Data Integration

- Combine data from multiple sources into a unified dataset.
- Example: Merging customer data from sales and marketing databases.

3. Data Transformation

- Convert data into suitable formats through normalization, aggregation, or encoding.
- Example: Scaling income values between 0 and 1.

4. Data Reduction

- Reduce dataset size while retaining important information.
- Techniques: Dimensionality reduction (PCA), attribute selection, sampling.
- Example: Selecting only relevant features for fraud detection.

5. Data Discretization

- Convert continuous attributes into categorical ones.

- Example: Converting age into ranges like "20–30," "30–40," etc.

DMBI 2023 May PYQ Answers

Q1. [5 Marks each]

- (A) Draw Data Warehousing Architecture.
- (B) What is **noisy data**? How to handle noisy data?
- (C) Compare and contrast between **OLTP and OLAP**.
- (D) Explain the concept of **information gain** and **Gini value** used in the decision tree algorithm.

Q2. [10 Marks each]

- (A) What is Data Mining? Explain the **KDD process** with a diagram.
- (B) Consider we have age of 29 participants in a survey given to us in sorted order.

5, 10, 13, 15, 16, 16, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70, 85 .

Compute the **mean, median, standard deviation, 1st and 3rd Quartile**.
Show the **Box and Whisker plot** for this data.

Q3. [10 Marks each]

- (A) Explain Market Basket Analysis with an example.
- (B) Consider Training dataset as given below. Use Naive Bayes Algorithm to determine whether it is advisable to play tennis on a day with hot temperature, rainy outlook, high humidity and no wind?

Outlook	temperature	Humidity	Windy	Class
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Play
rain	mild	high	false	Play
rain	cool	normal	false	Play
rain	cool	normal	true	No
overcast	cool	normal	true	Play
sunny	mild	high	false	No
sunny	cool	normal	false	Play
rain	mild	normal	false	Play
sunny	mild	normal	true	Play
overcast	mild	high	true	Play
overcast	hot	normal	false	Play
rain	mild	high	true	No

Q4. [10 Marks each]

(A) What is an outlier? Explain various methods for performing outlier analysis.

(B) Use the Apriori algorithm to identify the frequent item-sets in the following database. Then extract the strong association rules from these sets. Assume Min. Support = 50% Min. Confidence=75%

Tid	a	b	c	d	e	f	g
Items	1,2,4,5,6	2,3,5	1,2,4,5	1,2,4,5	1,2,3,4,5,6	2,3,4	1,2,4,5

Q5. [10 Marks each]

(A) Cluster the following eight points (with (x, y) representing locations) into three clusters:

A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9)

Assume Initial cluster centers are at: A1(2, 10), A4(5, 8) and A7(1, 2).

The distance function between two points $a = (x_1, y_1)$ and $b = (x_2, y_2)$ is defined as- $P(a, b) = |x_2 - x_1| + |y_2 - y_1|$

Use K-Means Algorithm to find the three cluster centres after the second iteration.

(B) Compare **Star schema**, **Snowflake schema**, and **Star Constellation**.

Q6. [10 Marks each]

Write short notes on:

(A) Dimensional Modeling.

(B) Random Forest Technique.

(C) Decision Tree Induction.

(D) Cross Validation.

(E) DBSCAN Algorithm.

Q1. [5 Marks each] - Answers

b. What is noisy data? How to handle noisy data?

Definition

Noisy data refers to **data containing errors, outliers, or random variations** that do not represent the true underlying pattern. It reduces the accuracy of analysis and decision-making.

Causes of Noisy Data

- Faulty sensors or instruments
- Human errors in data entry
- Transmission errors
- Irrelevant or redundant information

Methods to Handle Noisy Data

1. **Data Smoothing (Binning)** – Group values into bins and smooth by averaging or nearest neighbors.
2. **Regression Methods** – Fit data into a regression function to reduce noise.
3. **Clustering** – Detect and remove outliers by grouping similar data points.
4. **Data Cleaning** – Correct errors, remove duplicates, and standardize formats.
5. **Robust Algorithms** – Use models tolerant to noise (e.g., decision trees, ensemble methods).

d. Explain the concept of information gain and Gini value used in the decision tree algorithm.

Definition:

Information Gain and Gini Index are measures used in decision tree algorithms to choose the best attribute for splitting data.

Information Gain:

- Based on **entropy** (measure of randomness/impurity)
- Calculates how much uncertainty is reduced after a split
- Higher information gain → better attribute for splitting
- Used in algorithms like **ID3**

Formula concept:

Information Gain = Entropy (before split) – Entropy (after split)

Example:

If splitting data by "Outlook" reduces uncertainty in classifying "Play/No Play", it has high information gain.

Gini Index:

- Measures impurity of a dataset

- Value ranges from **0 (pure)** to **1 (impure)**
- Lower Gini value → better split
- Used in algorithms like **CART**

Formula concept:

$$\text{Gini} = 1 - \sum (\text{probability of each class})^2$$

Example:

If a node has mostly one class (e.g., mostly "Yes"), Gini value is low, indicating a good split.

Q2. [10 Marks each] - Answers

b. Consider we have age of 29 participants in a survey given to us in sorted order. 5, 10, 13, 15, 16, 16, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70, 85 .

Compute the mean, median, standard deviation, 1st and 3rd Quartile.

Show the Box and Whisker plot for this data.

Statistic	Value
Mean	30.69
Median	25
Std Deviation (population)	16.89
Q1	20
Q3	35.5
IQR	15.5
Lower Fence	5 (min, no low outliers)
Upper Fence	58.75
Outliers	70, 85

Working notes:

- $n = 29$ (odd), so median = 15th value = **25**
- $Q1 = \text{median of lower 14 values} = \text{avg of 7th \& 8th} = (20+20)/2 = \mathbf{20}$
- $Q3 = \text{median of upper 14 values} = \text{avg of 7th \& 8th} = (35+36)/2 = \mathbf{35.5}$



Q4. [10 Marks each] - Answers

b. Compare Star schema, Snowflake schema, and Star Constellation.

Feature	Star Schema	Snowflake Schema	Star Constellation (Galaxy Schema)
Structure	Simple star-like	Normalized, multi-level	Multiple fact tables with shared dimensions
Fact Tables	Single fact table	Single fact table	Multiple fact tables
Dimension Tables	Denormalized	Normalized into sub-tables	Shared across multiple fact tables
Complexity	Low	Medium to High	High

Feature	Star Schema	Snowflake Schema	Star Constellation (Galaxy Schema)
Query Performance	Fast (fewer joins)	Slower (more joins)	Moderate to complex
Redundancy	High	Low	Moderate
Storage Space	More	Less	Depends on design
Joins	Few joins	Many joins	Many joins (across facts)
Data Integrity	Lower	Higher	Moderate
Use Case	Simple reporting	Complex hierarchical data	Large enterprise systems
Example	Sales with Customer, Product	Customer split into City, State	Sales + Inventory sharing Product dimension

Q6. [5 Marks each] - Answers

a. Write short notes on: Dimensional Modeling.

Definition

Dimensional Modeling is a **design technique for data warehouses** that structures data into **facts and dimensions** to make it easy for users to query and analyze. It is widely used in **Business Intelligence (BI)** systems.

Key Points

- **Fact Table:**
 - Contains quantitative data (measures) such as sales, revenue, or quantity.
 - Linked to dimension tables via foreign keys.
- **Dimension Tables:**
 - Contain descriptive attributes (who, what, when, where, how).
 - Examples: Customer, Product, Time, Location.
- **Schemas Used:**
 - **Star Schema:** Central fact table connected directly to dimension tables.

- **Snowflake Schema:** Dimension tables normalized into multiple related tables.
- **Advantages:**
 - Simplifies complex queries.
 - Improves performance of OLAP operations.
 - Provides intuitive structure for business users.

Example

- **Retail Data Warehouse:**
 - **Fact_Sales Table:** Stores sales amount, units sold.
 - **Dimensions:**
 - Customer (Name, Age, Location)
 - Product (Category, Brand)
 - Time (Day, Month, Year)
 - Query: *"Total sales of electronics in Mumbai during March 2026."*

b. Write short notes on: Random Forest Technique.

Definition

Random Forest is an **ensemble learning technique** used for both classification and regression. It builds multiple decision trees and combines their outputs to improve accuracy and reduce overfitting.

Key Points

- **Working Principle:**
 - Creates many decision trees using random subsets of data and features.
 - Each tree votes (classification) or averages (regression).
 - Final prediction is based on majority vote or average result.

- **Features:**
 - Uses **bagging (bootstrap aggregating)** to train trees on different samples.
 - Introduces randomness in feature selection to ensure diversity among trees.
 - Reduces variance compared to a single decision tree.
- **Advantages:**
 - High accuracy and robustness.
 - Handles large datasets and missing values well.
 - Resistant to overfitting.
- **Limitations:**
 - Computationally expensive with many trees.
 - Less interpretable compared to a single decision tree.

Example

- **Bank Loan Prediction:**
 - Input features: Age, Income, Credit Score, Employment Status.
 - Random Forest builds multiple trees, each predicting whether a loan should be approved.
 - Final decision is based on majority voting across all trees.

c. Write short notes on: Decision Tree Induction

Definition

Decision Tree Induction is a **supervised learning method** used to build decision trees from labeled training data. It recursively splits the dataset into subsets based on attribute selection measures until pure class labels are reached.

Key Points

- **Process:**
 1. **Start with Training Data** – Contains attributes and class labels.

2. **Attribute Selection** – Choose the best attribute using measures like **Information Gain** or **Gini Index**.
 3. **Splitting** – Partition data based on attribute values.
 4. **Recursive Induction** – Repeat selection and splitting until nodes are pure or no attributes remain.
 5. **Leaf Nodes** – Represent final class labels.
- **Advantages:**
 - Simple and interpretable.
 - Handles categorical and numerical data.
 - Useful for classification tasks.
 - **Limitations:**
 - Can overfit if not pruned.
 - Sensitive to noisy data.

Example

- **Dataset:** Weather conditions → "Play Tennis?"
 - Attributes: Outlook, Temperature, Humidity, Wind.
 - Root Node: Outlook (highest Information Gain).
 - Sunny → split by Humidity.
 - Overcast → Play Tennis = Yes.
 - Rainy → split by Wind.

d. Write short notes on: Cross Validation

Definition

Cross Validation is a **model evaluation technique** used in machine learning to assess how well a model generalizes to unseen data. It helps avoid overfitting and ensures reliable performance estimation.

Key Points

- **Purpose:** To test the model's ability to predict new data not used during training.
- **Process:** Dataset is split into training and testing subsets multiple times.
- **Types of Cross Validation:**
 1. **k-Fold Cross Validation** – Dataset divided into k folds; model trained on $k-1$ folds and tested on the remaining fold. Repeated k times.
 2. **Leave-One-Out (LOO)** – Special case of k-fold where $k = n$ (each sample tested once).
 3. **Stratified k-Fold** – Ensures class distribution is preserved across folds (useful for imbalanced datasets).
 4. **Repeated Cross Validation** – k-fold repeated multiple times for more robust results.
- **Advantages:**
 - Provides a better estimate of model performance.
 - Reduces bias compared to a single train-test split.
 - Useful when dataset size is small.
- **Limitations:**
 - Computationally expensive for large datasets.
 - May still have variance depending on data splits.

Example

- Suppose we want to evaluate a **spam detection model**:
 - Using **5-fold cross validation**, the dataset is split into 5 parts.
 - The model is trained on 4 parts and tested on the 5th.
 - This process repeats 5 times, and the average accuracy across folds gives a reliable performance estimate.

DMBI 2023 December PYQ Answers

Q1. [5 Marks each]

(A) Explain the different types of attributes used in data exploration.

(B) Explain the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm with an example.

Q2. [10 Marks each]

(A) Explain the K-means algorithm in detail. Apply the algorithm to divide the following set of values into **3 clusters**: {2, 3, 6, 8, 9, 12, 15, 18, 22}.

(B) Compare Bagging and Boosting of a classifier

Q3. [10 Marks each]

(A) Explain **Multilevel and Multidimensional** Association rules with suitable examples.

(B) Using the given training dataset classify the following tuple using Naïve Bayes Algorithm: <Homeowner: No, Marital Status: Married, Job experience:3>

Homeowner	Marital Status	Job experience (in years)	Defaulted
Yes	Single	3	No
No	Married	4	No
No	Single	5	No
Yes	Married	4	No
No	Divorced	2	Yes
No	Married	4	No
Yes	Divorced	2	No
No	Married	3	Yes
No	Married	3	No
Yes	Single	2	Yes

Q4. [10 Marks each]

(A) Define data mining. Explain KDD process with help of a suitable diagram.

(B) For the table given perform Apriori algorithm and show frequent item set and strong association rules. Assume Minimum Support of 30% and Minimum confidence of 70%.

TID	Items
01	1, 3, 4, 6
02	2, 3, 5, 7
03	1, 2, 3, 5, 8
04	2, 5, 9, 10
05	1, 4

Q5. [10 Marks each]

(A) What is **noisy data**? How to handle it?

For the following data $D = \{4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34\}$ and Number of bins = 3

perform the following:

1. Partition into equal frequency bins.
2. Smoothing by bin means.
3. Smoothing by bin boundaries.

(B) Define data warehouse. Explain data warehouse architecture with help of a diagram.

Q6. [10 Marks each]

(A) What is an outlier? List types of outliers. Describe methods used for outlier analysis.

(B) Design BI system for Fraud Detection? Explain all steps from data collection to decision making.

Q2. [10 Marks each] - Answers

b. Compare Bagging and Boosting of a classifier

Feature	Bagging (Bootstrap Aggregating)	Boosting
Basic Idea	Train multiple models independently on random samples	Train models sequentially, focusing on errors
Training Style	Parallel	Sequential
Data Sampling	Random sampling with replacement	Weighted sampling (focus on misclassified data)
Goal	Reduce variance	Reduce bias and improve accuracy
Model Dependency	Independent models	Dependent models
Handling Errors	Does not focus on previous errors	Focuses on correcting previous errors
Overfitting	Reduces overfitting	Can overfit if noisy data
Sensitivity to Noise	Less sensitive	More sensitive
Weight Assignment	Equal weight to all models	Different weights to models
Example	Random Forest	AdaBoost